PF233-PB1R: PB1R transmit stage: gate drive, two 650 V FETs, shunt, flyback clamp, overcurrent comparator, 12 V aux

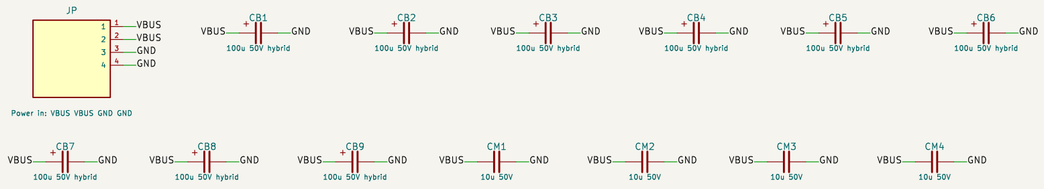
How to read: parts sit in functional blocks (dashed frames). Every pin ends in a short wire with its NET NAME; equal names are connected (also between blocks). A cross = intentionally unconnected pin.

Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

1 Power input and bulk bank

VBUS from the shaft (7S pack range 19.6 to 29.4 V) enters at JP (2 x VBUS, 2 x GND).

Nine 100 uF 50 V hybrid cans plus four 10 uF X7R cap the DC bus of the transmit stage.

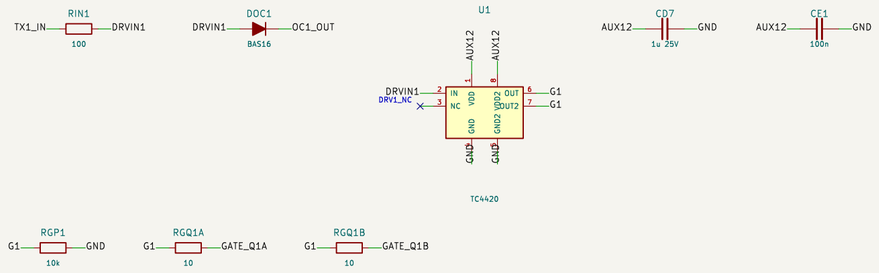


2 Gate drive

TX1_IN (logic from CB1D) -> RIN1 -> DRVIN1 -> U1 (TC4420, both halves paralleled) -> G1.

G1 -> RGQ1A / RGQ1B (10 R each) -> the gates of Q1A / Q1B; RGP1 pulls G1 low.

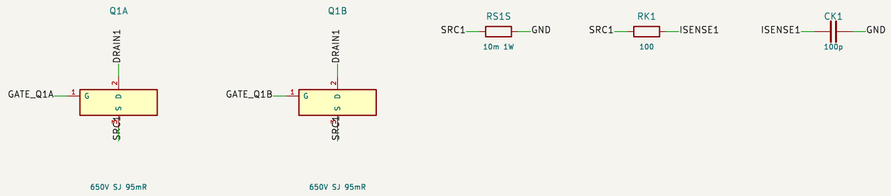
DOC1 can pull DRVIN1 low from the comparator output (overcurrent inhibit). U1 and the comparator run from AUX12.



3 Power switch and current shunt

Q1A, Q1B (650 V super-junction FETs, in parallel) switch DRAIN1 to SRC1.

RS1S 10 mohm shunt to GND; RK1 / CK1 take the Kelvin sense at SRC1 into ISENSE1.

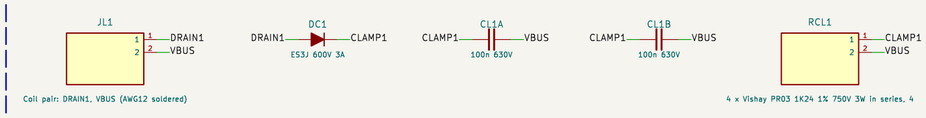


4 Coil connection and flyback clamp

JL1: coil outer band between DRAIN1 and VBUS.

DC1 (ultrafast diode) clamps the flyback from DRAIN1 into CLAMP1: CL1A / CL1B (630 V film / C0G) with the external bleed RCL1 (4 x Vishay PR03 1K24 in series, 4.96 k, potted in the NPM hot zone).

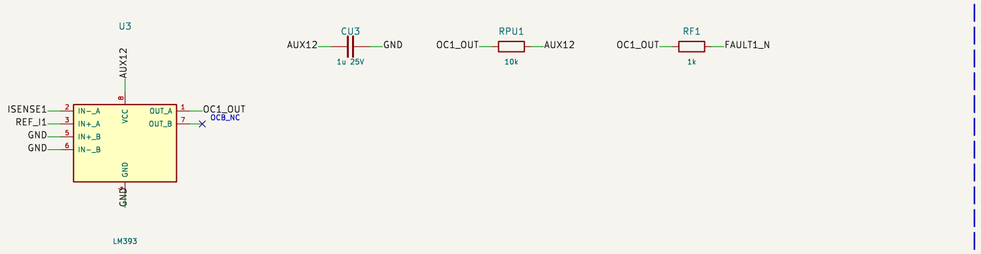
High-voltage nets DRAIN1 and CLAMP1: 1.0 mm clearance rule on the board.



5 Overcurrent comparator

U3 (LM393) compares ISENSE1 (IN-) with REF_I1 (IN+) from the controller: OC1_OUT is open collector, pulled up by RPU1 to AUX12.

RF1 -> FAULT1_N to CB1D: the same node drives DOC1 (gate-drive inhibit). The second comparator half has its inputs grounded and its output open (OCB_NC).



6 12 V aux supply

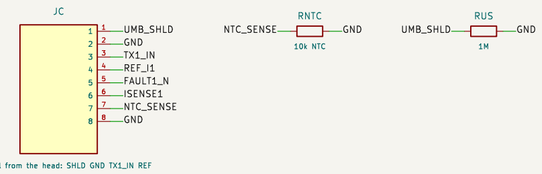
M12 (OKI-78SR-12 class module, datasheet unverified, ASSUMPTIONS 29) makes AUX12 from VBUS for the driver and comparator.



7 Umbilical to the head (JC) and sensing

JC pins: 1 SHLD, 2 GND, 3 TX1_IN, 4 REF_I1, 5 FAULT1_N, 6 ISENSE1, 7 NTC, 8 GND.

RNTC (10 k NTC on the FET spreader) -> NTC_SENSE; RUS bleeds the cable shield to GND.



Generated: src/pcb/sch_func.py; checked against the netlist.

MODELED design: not built, not bench-tested.

PB1R transmit stage: gate drive, two 650 V FETs, shunt, flyback clamp, overcurrent comparator, 12 V

ARX / PHI-FOLD 233

Sheet: /

File: PF233-PB1R_functional.kicad_sch

Title: PF233-PB1R functional schematic

Size: User	Date: 2026-10-26	**Rev: r3**
KiCad E.D.A. 9.0.9		Id: 1/1